

# playground

```
[1] "spec_tbl_df" "tbl_df"      "tbl"        "data.frame"
```

```
frog_names$scientificName[frog_names$scientificName == "Heleioporus australiacus  
australiacus"] <- "Heleioporus australiacus"
```

This line above might not have been the best approach, as some of the frogs are of different subspecies. For example: *Litoria verreauxii alpina* & *Litoria verreauxii verreauxii*

I can either:

1. Ignore subspecies for this project
2. Change the Scientific names in the `frog_names` dataset to remove subspecies

```
[1] 9158
```

```
[1] 9158
```

```
[1] 9158
```

```
[1] 0
```

```
[1] 0.06703019
```

But zooming out, we see that only 6.7% of our observations have missing commonNames. I say we proceed.

## Picking a Visualization Angle

There are a number of things we can choose to communicate:

1. Monthly/seasonal patterns of frog observations
2. Amount of frogs in each species observed
3. The range (distance) of frog observations
4. Regions where frogs were observed
5. When observations were made
6. How rare/abundant each frog is

And given that the data spans only 2023, we can use:

1. Monthly or seasonal patterns
2. Range of observations
3. Amount of observations
4. Regions of observations
5. When are the most observations made
6. Rarity

I would like to show rarities, and monthly or seasonal patterns on a map (range) for frog species

## Correcting Column Data Types

I want to create a new column called “Species”, taken from the last word in each observation’s Scientific Name. OR Factor the commonNames column

In addition to correcting the dtypes of other columns

```
spc_tbl_ [136,625 x 15] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ occurrenceID      : num [1:136625] 12832 12833 12834 12835 12836 ...
 $ eventID           : num [1:136625] 525618 526341 526673 526673 526673 ...
 $ decimalLatitude   : num [1:136625] -28.5 -33.7 -28.7 -28.7 -28.7 -30.4 -30.4 -30.4 ...
 $ decimalLongitude  : num [1:136625] 153 151 153 153 153 ...
 $ scientificName     : chr [1:136625] "Phyloria loveridgei" "Heleioporus australis" ...
 $ eventDate         : Date[1:136625], format: "2023-01-01" "2023-01-02" ...
 $ eventTime         : 'hms' num [1:136625] 11:18:32 20:39:30 21:30:07 21:30:07 ...
 ..- attr(*, "units")= chr "secs"
 $ timezone          : chr [1:136625] "GMT+1100" "GMT+1100" "GMT+1100" "GMT+1100" ...
 $ coordinateUncertaintyInMeters: num [1:136625] 10000 10000 10000 10000 10000 10000 10000 10000 ...
 $ recordedBy        : num [1:136625] 41480 834983 804177 804177 804177 ...
 $ stateProvince     : chr [1:136625] "New South Wales" "New South Wales" "New South Wales" ...
```

```

$ subfamily          : chr [1:136625] " Myobatrachid" NA " Myobatrachid" " Myobatrachid"
$ tribe              : chr [1:136625] " Limnodynastidae" NA " Myobatrachidae" " Myobatrachidae"
$ commonName         : chr [1:136625] "Masked Mountain Frog" NA "Giant Barred Frog" "Giant Barred Frog"
$ secondary_commonNames : chr [1:136625] "Loveridge's Mountain Frog" NA "-"
" "-" ...
- attr(*, "spec")=
.. cols(
..   occurrenceID = col_double(),
..   eventID = col_double(),
..   decimalLatitude = col_double(),
..   decimalLongitude = col_double(),
..   scientificName = col_character(),
..   eventDate = col_date(format = ""),
..   eventTime = col_time(format = ""),
..   timezone = col_character(),
..   coordinateUncertaintyInMeters = col_double(),
..   recordedBy = col_double(),
..   stateProvince = col_character()
.. )
- attr(*, "problems")=<externalptr>

tibble [136,625 x 15] (S3: tbl_df/tbl/data.frame)
 $ occurrenceID      : num [1:136625] 12832 12833 12834 12835 12836 ...
 $ eventID           : num [1:136625] 525618 526341 526673 526673 526673 ...
 $ decimalLatitude   : num [1:136625] -28.5 -33.7 -28.7 -28.7 -28.7 -30.4 -30.4 -30.4 ...
 $ decimalLongitude  : num [1:136625] 153 151 153 153 153 ...
 $ scientificName     : Factor w/ 186 levels "Adelotus brevis",...: 148 47 134 132 9 ...
 $ eventDate         : Date[1:136625], format: "2023-01-01" "2023-01-02" ...
 $ eventTime         : 'hms' num [1:136625] 11:18:32 20:39:30 21:30:07 21:30:07 ...
 ..- attr(*, "units")= chr "secs"
 $ timezone          : chr [1:136625] "GMT+1100" "GMT+1100" "GMT+1100" "GMT+1100" ...
 $ coordinateUncertaintyInMeters: num [1:136625] 10000 10000 10000 10000 10000 10000 10000 10000 ...
 $ recordedBy        : num [1:136625] 41480 834983 804177 804177 804177 ...
 $ stateProvince     : Factor w/ 9 levels "Australian Capital Territory",...: 2 2 2 2 2 ...
 $ subfamily         : Factor w/ 5 levels " Hylid"," Microhylidae",...: 3 NA 3 3 1 ...
 $ tribe             : Factor w/ 6 levels " Asterophryninae",...: 3 NA 4 4 5 4 4 5 ...
 $ commonName        : Factor w/ 179 levels "Atherton Tablelands Whirring Tree Frog",...: 179 NA 179 ...
 $ secondary_commonNames : Factor w/ 93 levels "-", "Australian Wood Frog",...: 48 NA 1 ...

```

## Exploratory Visualizations

Since we are visualizing monthly/seasonal variations, rarity, and range, I will visualize:

1. The monthly variations using the tribes and a faceted bar chart I would need to group by month (extracted from `eventDate`)
2. The rarity using all the tribes and a bar chart
3. The range using by calculating the minimum and maximum longitudes and latitudes to construct a square whose area I can calculate.

